

Computer Vision

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2. Robust Epipolar Geometry and Two-View Reconstruction

Essential matrix: $E = [t]_{\times} R$ with the relative rotation and translation computed from the two extrinsics as $R = R_2 R_1^T$ and $t = t_2 - R_2 R_1^T t_1$. Equivalently, using camera centers $C_i = -R_i^T t_i$, $t = R_2 (C_1 - C_2)$ and $E = [t]_{\times} R$.

Essential matrix

For calibrated cameras (normalized image coordinates), corresponding points x_1 and x_2 satisfy the epipolar constraint $x_2^T E x_1 = 0$, and the essential matrix is defined as $E = [t]_{\times} R$.

Here $[t]_{\times}$ is the 3×3 skew-symmetric matrix representing the cross product with t .

Relative pose from P_1 and P_2

Given $P_1 = [R_1 \ t_1]$ and $P_2 = [R_2 \ t_2]$ (world $\rightarrow$ camera mappings), the relative motion from camera 1 to camera 2 is $R = R_2 R_1^T$ and $t = t_2 - R_2 R_1^T t_1$.

Using the relation between a camera center and its extrinsics, $C = -R^T t$, this can also be written as $t = R_2 (C_1 - C_2)$.

Derivation outline

Start from the rigid transforms $X_1 = R_1 X + t_1$ and $X_2 = R_2 X + t_2$; eliminating X gives $X_2 = (R_2 R_1^T) X_1 + (t_2 - R_2 R_1^T t_1)$, so R and t are the relative pose parameters.

Substituting these into the calibrated epipolar form yields $E = [t]_{\times} R$, which enforces $x_2^T E x_1 = 0$.

Essential matrix E has 5 degrees of freedom; the minimal number of point correspondences to determine E is 5; the eight-point algorithm uses 8 correspondences; with 15% outliers and 8-point samples, about 15 RANSAC iterations are needed for 99% success.

Degrees of freedom

An essential matrix can be written as $E = [t]_{\times}R$, where R (3 DOF) is rotation and t (2 effective DOF after fixing scale) is translation direction, giving 5 DOF when considered up to scale.

Minimal correspondences

Each point correspondence supplies one scalar epipolar constraint, so at least 5 points are required to solve for the 5-DOF essential matrix in a minimal way (the “five-point algorithm”).

Eight-point algorithm

The classical eight-point algorithm solves a homogeneous linear system in the 9 entries of E (or F) with a rank-8 constraint, so it requires at least 8 point correspondences.

RANSAC iterations

For RANSAC, the required number of iterations is $N = \frac{\log(1-p)}{\log(1-w^s)}$, where p is desired success probability, w the inlier fraction, and s the sample size. Here $p = 0.99$, outlier fraction = 0.15 $\Rightarrow w = 0.85$, and $s = 8$, giving $N \approx 14.5$, so one uses 15 iterations.

3. Levenberg-Marquardt for Structure from Motion Problems

The Jacobian $J_i(X_j)$ has the given 2×4 form because each residual component is a ratio of linear functions of X_j , and stacking all camera residuals simply concatenates these 2-vectors and 2×4 blocks. The stacked residual for point X_j has size $2m$ and the stacked Jacobian has size $2m \times 4$.

Setup and notation

For camera i , introduce $Z_{ij} = P_i X_j = (Z_1, Z_2, Z_3)^T$, so the projected point is $\hat{x}_{ij,1} = Z_1/Z_3$, $\hat{x}_{ij,2} = Z_2/Z_3$.

Then the residuals are $r_{i,1}(X_j) = x_{ij,1} - Z_1/Z_3$ and $r_{i,2}(X_j) = x_{ij,2} - Z_2/Z_3$.

Part I: Jacobian $J_i(X_j)$

First differentiate r_i with respect to Z_{ij} :

For $r_{i,1}$: $\partial(Z_1/Z_3)/\partial Z = (1/Z_3, 0, -Z_1/Z_3^2)$, hence

$$\frac{\partial r_{i,1}}{\partial Z_{ij}} = \left(-\frac{1}{Z_3}, 0, \frac{Z_1}{Z_3^2}\right).$$

For $r_{i,2}$: $\partial(Z_2/Z_3)/\partial Z = (0, 1/Z_3, -Z_2/Z_3^2)$, hence

$$\frac{\partial r_{i,2}}{\partial Z_{ij}} = \left(0, -\frac{1}{Z_3}, \frac{Z_2}{Z_3^2}\right).$$

Since $Z_{ij} = P_i X_j$ is linear, $\partial Z_{ij}/\partial X_j = P_i$ (a 3×4 matrix), so by the chain rule

$$J_i(X_j) = \frac{\partial r_i}{\partial X_j} = \begin{bmatrix} \frac{\partial r_{i,1}}{\partial Z_{ij}} \\ \frac{\partial r_{i,2}}{\partial Z_{ij}} \end{bmatrix} P_i.$$

Writing P_i by rows P_i^1, P_i^2, P_i^3 and using $Z_1 = P_i^1 X_j$, $Z_3 = P_i^3 X_j$ gives for the first row

$$\frac{\partial r_{i,1}}{\partial X_j} = -\frac{1}{Z_3} P_i^1 + \frac{Z_1}{Z_3^2} P_i^3 = \frac{P_i^1 X_j}{(P_i^3 X_j)^2} P_i^3 - \frac{1}{P_i^3 X_j} P_i^1,$$

and similarly for the second row with $P_i^2 X_j$.

Thus

$$J_i(X_j) = \begin{bmatrix} \frac{P_i^1 X_j}{(P_i^3 X_j)^2} P_i^3 - \frac{1}{P_i^3 X_j} P_i^1 \\ \frac{P_i^2 X_j}{(P_i^3 X_j)^2} P_i^3 - \frac{1}{P_i^3 X_j} P_i^2 \end{bmatrix},$$

which is the required 2×4 Jacobian used in bundle adjustment.

Part II: Stacked residuals and Jacobian

Define the stacked residual for point X_j by concatenation

$$r(X_j) = \begin{bmatrix} r_1(X_j) \\ r_2(X_j) \\ \vdots \\ r_m(X_j) \end{bmatrix} \in \mathbb{R}^{2m}, J(X_j) = \begin{bmatrix} J_1(X_j) \\ J_2(X_j) \\ \vdots \\ J_m(X_j) \end{bmatrix} \in \mathbb{R}^{2m \times 4}.$$

Then

$$r(X_j) + J(X_j) \delta X_j = \begin{bmatrix} r_1(X_j) + J_1(X_j) \delta X_j \\ \vdots \\ r_m(X_j) + J_m(X_j) \delta X_j \end{bmatrix},$$

so its squared norm is exactly the sum of the squared norms of the blocks,

$$\| r(X_j) + J(X_j) \delta X_j \|^2 = \sum_{i=1}^m \| r_i(X_j) + J_i(X_j) \delta X_j \|^2,$$

as required.